Yong Li Neo

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EDUCATION

Georgia Institute of Technology (GT), Atlanta, GA December 2027
Bachelor of Science in Computer Science (GPA: 4.0)

SKILLS & INTEREST

Programming Languages: C, C++, Python, Java, R
ML/AI Frameworks & Tools: TensorFlow, PyTorch, NumPy, Git, SwiftUI, SciKit-learn
Languages: English (native), Chinese (native), Malay (Fluent)
Interest: Artificial Intelligence, Machine Learning, Deep Learning, Cell-Cell Interactions

WORK EXPERIENCE

Research Assistant, Georgia Institute of Technology, Atlanta, GA January 2026 - Present

- Worked under Prof. Xiuwei Zhang in Zhang CompBio group as an undergraduate researcher
- Performed work on discovery & modeling of cell-cell interactions using Graph Neural Networks
- Participated in weekly meetings to update research outcomes & discussion with peers within group

RAE Architect, *Intern*, Malacca, Malaysia Summer 2024

- Coordinated project documentation and optimized hospital development planning by identifying blockers and implemented solutions
- Drafted contracts and participated in client meetings
- Automated workflow-procedure reducing time by **8%** by writing a Python script

RESEARCH & PUBLICATIONS

Depression Detection Machine Learning Model

- Designed and optimized machine learning models to detect depression from EEG data
- Achieved 80% accuracy in identifying depressed patients in collaboration with a local mental health clinic
- Benchmarked Elastic Net, SVM, Random Forest, XGBoost, CatBoost to compare efficiency for medical signal processing
- **Publication: Y. L. Neo**, "A Review of State-of-the-Art Machine Algorithms on Classifying Depressive Disorder EEG Signals," 2024 ICAICA (IEEE conference), Dalian, China

Deep Learning Model for Confusion Signals Detection

- Implemented CNN-biLSTM, CNN-biGRU, Bi-RNN, Bi-GRU architectures to classify EEG signals
- Obtained an accuracy of 76.7% for the most optimal model
- **Publication:** Z. Y. Lim and **Y. L. Neo**, "Confused vs Non-Confused Electroencephalography Signal Classification Using Deep Learning Algorithm," 2023 I2CACIS (IEEE conference), Shah Alam, Malaysia

Seizure Detection Machine Learning Model

- Developed and validated seizure detection models using EEG datasets
- Improving detection reliability with Extra Trees, Naïve Bayes, SGD, and Logistic Regression, achieving 96.6% accuracy
- **Publication:** Z. Y. Lim and **Y. L. Neo**, "Epileptic Seizure Electroencephalography Signal Classification Using Machine Learning Algorithm," 2022 CSCI (IEEE conference), Las Vegas, NV, USA

LEADERSHIP

Malaysian Informatics and Programming Society (MIPS), *National Trainer & Problem Setter* December 2023 – May 2025

- Developed lectures, weekly online workshops and materials for the top 50 national finalists
- Designed challenges for national-level Informatics Olympiads rounds: Dynamic Programming, Binary Search and Sorting algorithms

Robotics Club & Tech Club (KTJ), *Vice President & Head of Teaching* March 2024 – May 2025

- Led 30+ weekly meetings to discuss and plan lectures/projects
- Mentored 60 junior members in Arduino and C++ basics

HONORS & AWARDS

- President's Undergraduate Research Awards (PURA Salary Awards)
- International Olympiad in Informatics (IOI), Participation Award
- Asia Pacific Olympiad in Informatics (APIO 2023, 2024), Participation Award
- Malaysian Computing Olympiad (Silver & Bronze Medals)
- Malaysian Computing Challenge (Gold, Silver & Bronze Medals)
- Kolej Tuanku Ja'afar Scholarship, Recipient